Shangyuan Ye

CONTACT INFORMATION

Affiliation: Assistant Professor, Department of Mathematics and Statistics, Florida International University
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EDUCATION

Oklahoma State University Stillwater, OK
Ph.D., Statistics May 2018
Dissertation: Bayesian Analysis for Sparse Functional Date
Advisor: Dr. Ye Liang, Ph.D.

University of Texas at Dallas Richardson, TX
B.A., Mathematics, concentration in Statistics May 2014
Project: Limited-Fluctuation Credibility with Uncertain Priors
Advisor: Dr. Michael Baron, Ph.D.

RESEARCH EXPERIENCE

Florida International University, Department of Mathematics and Statistics, Miami, FL
Assistant Professor 8/2025-Present

Oregon Health & Science University, Biostatistics Shared Resource, Knight Cancer Institute, Portland, OR
Assistant Staff Scientist 7/2022-8/2025

Oregon Health & Science University, School of Medicine, Portland, OR
Affiliated Member 9/2022-Present

Oregon Health & Science University, Biostatistics Shared Resource, Knight Cancer Institute, Portland, OR
Biostatistician 3 6/2021-7/2022

Harvard Pilgrim Health Care Institute and Harvard Medical School, Department of Population Medicine, Boston, MA
Research Fellow 10/2019-5/2021
Advisor: Dr. Rui Wang, Ph.D.

University of Massachusetts Medical School, Department of Population and Quantitative Health Sciences, Worcester, MA
Post Doctoral Associate 6/2018-10/2019
Advisor: Dr. Bo Zhang, Ph.D.

HONORS and AWARDS

Knight QOP Pilot Award: Adding climate-related exposures and neighborhood socioeconomic context to HOP, 01/2025–12/2025. Biostatistician. (PI: Charlotte Roscoe)

Knight CPC Pilot Award: Investigating interval breast cancer as a distinct molecular entity associated with increased aggressiveness, 09/2024–08/2025. Biostatistician. (PI: Zhenzhen Zhang)

OHSU Talwalkar Grant: Reducing Asparaginase Complications in Leukemia, 09/2024–12/2025. Biostatistician. (PI: Lori E Morgan)

DoD Grant: Impact of recent childbirth on breast cancer prognosis and tumor molecular phenotypes among young Black and White women, 05/2024–04/2027. Biostatistician (PI: Pepper J Schedin)

DoD Grant: Characterizing prostate cancer progression through integrated analysis of single-cell atlas, 07/2024–06/2028. Co-Investigator (PI: Zheng Xia)

Travel Award for Advances in Statistical and Computational Methods for Analysis of Biomedical, Genetic, and Omics Data, 2023

Knight CCSG Award: Impact of Pregnancy on Survival in Young Women Breast Cancer Patients Using the Manchester Cohort Study. 12/2021–11/2022. Biostatistician. (PI: Zhenzhen Zhang)

Harvard Pilgrim Health Care Institute and Harvard Medical School Thomas O. Pyle Fellowship, 07/2020–07/2021

Oklahoma State University Department of Statistics Student Travel Award, 2018

PUBLICATIONS: STATISTICAL METHODOLOGY AND APPLICATION

Ye, S., Rakshe, S., Liang, Y. Model-free variable selection and hypothesis testing for high-dimensional data analysis. arXiv preprint arXiv:2410.19031. In revision for Journal of the American Statistical Association

Ye, S., Cruz, M., Wang, Z., Yun, Y. Marginalized zero-one inflated Beta regression models for analyzing Interrupted time series analysis with proportional outcomes. arXiv preprint arXiv:2212.09996. In revision for Journal of Applied Statistics

Ye, S., Yu, T., Caroff, D. A., Huang, S. S., Zhang, B., Wang R. Variable selection in modelling clustered data via withincluster resampling. *Canadian Journal of Statistics* 2025; 53(1): e11824

Yu, T., **Ye, S.**, Wang, R. High-dimensional variable selection accounting for heterogeneity in regression coefficients across multiple data sources. *Canadian Journal of Statistics* 2024; 52(3): 900–923.

Ye, S., Li, D., Yu, T., Caroff, D. A., Guy, J., Poland, R. E., Sands, K. E., Septimus, E. J., Huang, S. S., Platt, R., Wang, R. The impact of surgical volume on hospital ranking using the standardized infection ratio. *Scientific Reports* 2023; 13(1).

Ye, S., Wang, R., Zhang, B. Comparison of estimation methods and sample size calculation for parameter-driven interrupted time series models with count outcomes. *Health Services and Outcomes Research Methodology* 2022; 19: 1-48

Ye, S., Liang, Y., Zhang, B. Bayesian functional mixed-effects models with grouped smoothness for analyzing time-course gene expression data. *Current Bioinformatics* 2021; 16(1): 2–12.

Baron, M., **Ye, S.**, Chattopadhyay, B. Limited-fluctuation credibility with uncertain priors. *Variance* 2020; 14(1).

Liu, W., Haran, J.P., **Ye, S.**, Tjia, J., Bucci, V., Zhang, B. High-dimensional causal mediation analysis with a large number of mediators clumping at zero at assess the contribution of the microbiome to the risk of bacterial pathogen colonization in order adults. *Current Bioinformatics* 2020; 15(7): 671–696

Liu, W., **Ye, S.**, Barton, B., Fischer, M.A., Lawrence, C., Rahn, E.J., Danila, M.I., Saag, K.G., Harris, P.A., Lemon, S.C., Allison, J.J., Zhang, B. Simulation-based power and sample size calculation for designing interrupted time series analyses of count data outcomes in evaluation of health policy interventions. *Contemporary Clinical Trials Communications* 2020; 17: 100474

Zhang, B., **Ye, S.**, Shankara, S.B., Zhang, H., Zheng, Q. Neutral diagnosis: An innovative concept for medical device clinical trials. *Contemporary Clinical Trials Communications* 2019; 16: 100436

Ye, S., Liang, Y., Ahmad, A.I. Orthogonal series density estimation for complex surveys. *Journal of Nonparametric Statistics* 2019; 31 (2): 1–13.

PUBLICATIONS: COLLABORATION AND REVIEW

Kumaraswamy, A., Hu, Y.M., Yates, J., Zhang, C., **Ye, S.**, Ryan, C.J., Quigley, D., Aggarwal, R.R., Reiter, R.E., Beer, T.M., Rettig, M., Gleave, M., Lara, P.N., Stuart, J., Thomas, G.V., Feng, F.Y., Small, E.J., Xia, Z., Alumkal, J.J. Transcriptional profiling to identify a program of enzalutamide extreme non-response in lethal prostate cancer. *npj Precision Oncology* 2025

Lee, S.W., Ekstrom, T., Manalo, E.C., **Ye, S.**, Berry, M., Grygoryev, D., Szczepaniak, M., Lee, J., Marmolejo, C.O., Haverlack, S., Lee, J. HMG Box-Containing Protein 1 (HBP1) Protects Against Pancreatic Injury in Acute Pancreatitis but Promotes Neoplastic Progression. *Cellular and Molecular Gastroenterology and Hepatology* 2025; 19(9)

Zhou, P., Li, Z., Liu, F., Kwon, E., Hsieh, T. C., **Ye, S.**, Vasudevan, S., Lee, J. and Zhou, C. BAMBI: Integrative biostatistical and artificial-intelligence method discover coding and non-coding RNA genes as biomarkers. *Briefings in Bioinformatics* 2025; 26(2)

Togioka, B. M., Rakshe, S. K., **Ye, S.**, Tekkali, P., Tsikitis, V., Fang, S. H., Herizig, D. O., Lu, K. C. and Aziz, M. F. A Randomized Controlled Trial of Sugammadex vs. Neostigmine and Glycopyrrolate for Reversal of Rocuronium Blockade on Gastric Emptying in Adults Undergoing Elective Colorectal Surgery. *Anesthesia & Analgesia* 2025

Harrar, D. B., Salussolia, C. L., Keith, L. G., **Ye, S.**, Meadows, J. S., Fialkow, A., Zhu, G., VanderPluym, C., Zhang, B. Rivkin, M. J. Development and Validation of a Postprocedure Stroke Screening Tool in Children With Cardiac Disease. *Journal of Child Neurology* 2025; 40(4): 264–272

Obeidat, S., Rakshe, S., **Ye, S.**, Cassese, B., Clancy, L., Togioka, B. Impact of Unconscious Race Bias Amongst Anesthesia Providers on Nonverbal Communication During the Presurgical Anesthesia Consult: A Prospective, Observational Cohort Study. *Anesthesia & Analgesia* 2024; 139(5): 789–792

Taflin, N., Sandow, L., Thawani, R., **Ye, S.**, Kardosh, A., Corless, C.L., Chen, E.Y.. Survival in metastatic microsatellite-stable colorectal cancer correlated with tumor mutation burden and mutations identified by next-generation sequencing. Journal of Gastrointestinal Oncology 2024; 15(2): 681.

Zhang, Z., **Ye, S.**, Bernhardt, S. M., Nelson, H. D., Velie, E. M., Borges, V. F., Woodward, E. R., Evans, D. G. R., Schedin, P. J. Postpartum Breast Cancer and Survival in Women With Germline BRCA Pathogenic Variants. *JAMA Network Open* 2024; 15(4): e247421

Togioka, B. M., Harriman, K. A., **Ye, S.**, Berli, J. Frequency and Characteristics of Postoperative Neuropathy in Individuals on Gender-Affirming Hormone Therapy Undergoing Gender Affirmation Surgery: A Retrospective Cohort Study. *Cureus* 2023; 15(10)

Biebelberg, B., **Ye, S.**, Wang, R., Klompas, M., Rhee, C. Association Between Negative Pressure Room Utilization and Hospital-Acquired Aspergillus Rates in Patients with COVID-19 in Two Academic Hospitals. *Infection Control & Hospital Epidemiology* 2023; 44(12): 2085–2088

Ye, S., Lim, J., Huang, W. Statistical considerations for repeatability and reproducibility of quantitative imaging biomarkers. *The British Institute of Radiology Open* 2022; 4(1)

Hayes, M., Yu, Y., Bassale, S., Chakiryan, N., Chen, Y., **Ye, S.**, Garzotto, M., Kopp, R. Calibrated regression models based on the risk of clinical nodal metastasis should be used as decision aids for prostate cancer staging to reduce unnecessary imaging. *Clinical Genitourinary Cancer* 2022; 20(6): 490–497

Klompas, M., **Ye, S.**, Vaidya, V., Ochoa, A., Baker, M. A., Hopcia, K., Hashimoto, D., Wang, R., Rhee, C. Association between Airborne Infection Isolation Room Utilization Rates and Healthcare Worker COVID-19 Infections in Two Academic Hospitals. *Clinical Infectious Diseases* 2022: 74(12): 2230–2233

Winer, K. K., **Ye, S.**, Ferre, E. M. N., Schmitt, M. M., Zhang, B., Heller, T., Cutler, G. B., Lionakis, M. S. Therapy with PTH 1-34 or Calcitriol and Calcium in Diverse Etiologies of Hypoparathyroidism over 27 Years at a Single Tertiary Care Center. *Bone* 2021; 149

Rhee, C., Wang, R., **Ye, S.**, Baker, M., Griesbach, D., Laskowski, K., Klompas, M. Decline in SARS-CoV-2 Infection Rates Among Healthcare Workers in Two Hospitals Following Widespread Administration of mRNA Vaccines. *Open Forum Infectious Diseases* 2021; 8(8)

Yang, Y., Lu, L., Chen, T., **Ye, S.**, Kelifa, M. O., Zhang, J., Liang, T., Hu, B., Gao, X., Wang, W. Healthcare workers mental health and their associated predictors during the epidemic peak of COVID-19. *Psychology Research and Behavior Management* 2021; 14: 221–231

Kang, L., **Ye, S.**, Zhang, N., Zhang, B. A segment logistic regression approach to evaluating change in Caesarean section rate with reform of birth planning policy in two regions in China from 2012 to 2016. *Risk Management and Healthcare Policy* 2020; 13: 245–253

Ye, S., Zhang, H., Shi, F., Guo, J., Wang, S., Zhang, B. Ensemble methods to improve prediction accuracy of suspected fetal macrosomia and large-for-gestational age. *Journal of Clinical Medicine* 2020; 9(2): 380

Liu, Y., **Ye, S.**, Xiao, X., Sun, C., Wang, G., Zhang, B., Wang, G. Tuning, selection, and ensemble of multiple risk scores for predicting type 2 diabetes using machine learning techniques. *Risk Management and Healthcare Policy* 2019; 12: 189–198

Zhang, B., Shankara, S.B., **Ye, S.**, Zhang, H. Design and analysis of high-risk medical device clinical trials for diabetes monitoring and treatment: A review. *Journal of Pancreatology* 2019; 2(4): 123–130

Kang, L., Gu, H., **Ye, S.**, Xu, B., Jing, K., Zhang, N., Zhang, B. Rural-urban disparities in caesarean section rates in minority areas in China: Evidence from electronic health records. *Journal of International Medical Research* 2019; 48(2)

Liu, Y., **Ye, S.**, Xiao, X., Zhou, T., Yang, S., Wang, G., Sun, C., Zhang, B., Wang, G. Association of diabetes mellitus with hepatitis B and hepatitis C virus infection: Evidence from an epidemiology study. *Infection and Drug Resistance* 2019; 12: 2875–2883

MANUSCRIPTS IN PREPARATION

Ye, S. Statistical inference for high-dimensional clustered data via partial sufficient association. In progress

Ye, S., Yu, T., Wang, R. Conditional correlation between hospital ranking systems accounting for estimation uncertainty in rankings. In progress

Mostafa, S. A., **Ye, S.** Orthogonal Series Regression Estimation for Complex Surveys. In progress

CONFERENCE PRESENTATIONS

"Statistical inference for high-dimensional clustered data via partial sufficient dimension reduction," WNAR annual meeting, Whistler, British Columbia, Invited Presentation, 2025

"Model-free Inference for High-dimensional Data," ASA Oregon Chapter Meeting, Corvallis, Oregon, Invited Presentation, 2024

"Model-free variable selection and hypothesis testing for high-dimensional data analysis," Joint Statistical Meetings, Portland, Oregon, Contributed Presentation, 2024

"Model-free variable selection and hypothesis testing for high-dimensional data analysis," WNAR annual meeting, Fort Collins, Colorado, Contributed Presentation, 2024

"Statistical Literacy for Medical Literature", Department of Anesthesiology and Perioperative Medicine, Oregon Health & Science University, Oregon, Guest Lecture, 2024

"Power and sample size calculation in grant writing", Department of Bioengineering, Oregon Health & Science University, Oregon, Guest Lecture, 2024

"Variable selection in modeling cluster data via within-cluster resampling," WNAR annual meeting, Anchorage, Alaska, Contributed Poster Presentation, 2023

"Marginalized Three-Part Beta Interrupted Time Series Regression Models for Proportional Data Analysis," 6^{th} Stat4Onc Annual Symposium, Portland, Oregon, Contributed Poster Presentation, 2023

"Variable selection in modeling cluster data via within-cluster resampling," Advances in Statistical and Computational Methods for Analysis of Biomedical, Genetic, and Omics Data, Richardson, Texas, Contributed Poster Presentation, 2023

"Marginalized Three-Part Beta Interrupted Time Series Regression Models for Proportional Data Analysis," Joint Statistical Meetings, Washington D.C., Contributed Presentation, 2022

"Power and sample size calculation for interrupted time series analyses of count outcomes," WNAR annual meeting, Virtual Conference, Invited Presentation, 2021

"Orthogonal series function estimation," Department of Biostatistics, Harvard T. H. Chan School of Public Health, Boston, Massachusetts, Guest Lecture, 2021

"Variable selection in modeling cluster data via within-cluster resampling," Worcester State University, Worcester, Massachusetts, Invited Presentation, 2020

"Simulation-Based Power and Sample Size Calculation for Designing Interrupted Time Series Analyses of Count Outcomes in Evaluation of Health Policy Interventions," Joint Statistical Meetings, Virtual Conference, Contributed Presentation, 2020

"Orthogonal series density estimation for complex surveys," Joint Statistical Meetings, Baltimore, Maryland, Contributed Presentation, 2017

"Limited-Fluctuation Credibility with Uncertain Priors," ASA Oklahoma Chapter Meeting, Oklahoma City, Oklahoma, Poster Presentation, 2016

SERVICES

Reviewers
Annals of Applied Statistics, Statistics in Medicine, Journal of Statistical Computation and Simulation, Contemporary Clinical Trials, Health Services and Outcomes Research Methodology, Mathematics, Entropy, Genes, Cancers, Algorithms

Institutional Services
OHSU Data & Safety Monitoring Committee, 2023-Present
Organizer, OHSU Biostatistics Shared Resource Education Series, 2024-Present
Organizer, OHSU Biostatistics Shared Resource Journal Club, 2022-Present
OHSU Clinical Trials Training Workshop Committee, 2023-2024

TEACHING AND MENTORING

Oregon Health & Science University
Lecturer, *Statistical Inference II,* Spring 2024
Lecturer, *Statistical Inference I,* Winter 2022, Winter 2023, Winter 2024, Winter 2025
Lecturer, *Introduction of Biostatistics,* Winter 2023, Winter 2024, Winter 2025
Lecturer, *Introduction to Probability,* Fall 2022
Lecturer, *Categorical Data Analysis,* Spring 2022

Oklahoma State University
Lecturer, *Elementary Statistics,* Summer 2016, Fall 2016, Fall 2017, Spring 2018
Lecturer, *Elementary Statistics for Business and Economics,* Summer 2017
Lecturer, *Engineering Statistics with Design Experiments,* Spring 2017
Graduate Teaching Assistant, *Categorical Data Analysis,* Spring 2016
Graduate Teaching Assistant, *SAS Programming,* Spring 2016
Graduate Teaching Assistant, *Mathematical Statistics,* Spring 2015, Fall 2015, Spring 2016
Graduate Teaching Assistant, *Elementary Statistics & Elementary Statistics for Business and Economics,* 2014-2016